



## MANIPAL ACADEMY OF HIGHER EDUCATION

### THIRD SEMESTER B.TECH. (COMPUTER SCIENCE & ENGINEERING) EXAMINATIONS - DECEMBER 2022

#### SUBJECT : CSE 2151 - COMPUTER ORGANIZATION AND ARCHITECTURE COMPUTER ORGANIZATION AND ARCHITECTURE [CSE 2151]

Marks: 50

Duration: 180 mins.

#### A

Answer all the questions.

- 1A) Represent the following integers in 2's complement representation with minimum number of bits required, perform the operation mentioned and state whether overflow occurs or not in each case. Also, if overflow has occurred, state why it has occurred. (3)
- i) Add +8 with +8
  - ii) Subtract -11 from -15
- 1B) Assume that an array A is stored in consecutive word locations starting from address A and an array B is stored in consecutive word locations starting from address B. Write a RISC-style program to find the product of the corresponding elements of two arrays and store it in the third array C starting at address C. Assume uniform sized arrays with the size of the array stored in memory location named SIZE. Assume a byte addressable memory and word length of two bytes. (3)
- 1C) Consider two decimal numbers  $A = +89.125$   $B = +15.5$   
Perform the following operations: (4)
- a) Convert the decimal numbers to floating point binary.
  - b) Normalize the number obtained above in (a).
  - c) Represent the normalized numbers in IEEE 32-bit floating point representation.
  - d) Perform  $(A / B)$ , on the numbers represented in (c) using the rules for division of IEEE 32-bit floating point representation. Indicate all the steps clearly.
  - e) Store the result in the IEEE 32-bit floating point representation.
- 2A) Perform the following arithmetic operation using the rules for arithmetic operations in IEEE single precision format: (3)
- $-116.5 - (+211.75)$
- 2B) Make use of paper and pencil method to show the division of 22 by 4 in its binary representation. Convert the final answer to its corresponding decimal representation. Show all the necessary steps. (2)
- 2C) Design a Register Transfer Description to find  $x^y$ . Validate the Register Transfer Description and identify the hardware elements to be used in the Processing Unit. (5)
- 3A) Design and draw a microprogrammed control organization for the Register Transfer Description given below with necessary details. Show all the steps in details. (3)

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        Declare registers A[8], B[8], C[8], I[8];
        Declare inbus[8], outbus[8];
Start:   A ← inbus, I ← 3;
        B ← inbus;
Back:   C ← A+B;
        If C[0]=0 then go to Deca;
        If C[0]=1 then go to Decb;
Deca:   A ← A-1;
        go to Next;
Decb:   B ← B-1;
Next:   I ← I-1;
        If I > 0 then go to Back;
Halt:   go to Halt

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- 3B) A computer has a small data cache capable of holding eight words and has a byte addressable memory. Each cache block consists of one word and one word comprises of one byte. When a given program is executed, the processor reads data sequentially from the addresses given below (Addresses are given in decimal). Assume that the cache is initially empty. Show the contents of the cache after each memory access, for a Set associative mapped cache that uses the LRU replacement algorithm and compute the hit rate, if the set comprises of 2 blocks. (5)
- 7200, 7201, 7208, 7213, 7200, 7216, 7208, 7201, 7215, 7201, 7213, 7218, 7220, 7213, 7217, 7219
- 3C) Consider a long sequence of accesses to a disk with an average seek time of 7 ms and an average rotational delay of 4 ms. The average size of a block being accessed is 8K bytes. The data transfer rate from the disk is 34 Mbytes/sec. Assuming that the data blocks are randomly located on the disk, estimate the portion of time occupied by seek and rotational delay for the situation in which disk accesses are arranged so that in 90 percent of the cases, the next access will be to a data block on the same cylinder. (2)
- 4A) With a diagrammatic illustration, discuss the Organization of a 2Mx32 memory module using 512Kx8 static memory chips. Also, explain its working. (2)
- 4B) Consider a fully associative cache with 8 cache blocks (0-7). The memory block requests are in the order:  
4, 3, 25, 8, 19, 6, 25, 8, 16, 35, 45, 22, 8, 3, 16, 25, 7  
Outline the steps to demonstrate the working of LRU with a diagrammatic illustration for the memory block requests given above. If LRU replacement policy is used, which cache block will have memory block 7? Also, calculate the hit ratio and miss ratio. (4)
- 4C) What is an interrupt? Explain the Concept of Vectored Interrupts and Interrupt Nesting in the context of Handling Multiple Devices. (4)
- 5A) Compare and Contrast the I/O Program Controlled Transfer with DMA Transfer. (2)
- 5B) What is Instruction Pipeline? Interpret the effect of conditional branch on instruction pipelining with a Timing Diagram. (3)
- 5C) Summarize the principle involved in write-through protocol and write-back protocol. Explain the concept of Transfer of Control through the use of interrupts with a diagram. (5)

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